StudentShield: design flowchart

Everything below runs on IBM LinuxONE (s390x). Top: building the model in Jupyter. Bottom: what happens live when each payment lands.

1 BUILD THE MODEL · Jupyter container on LinuxONE (notebooks 01–03)

IBM AML HI-Small

5.08M synthetic payments
1 in 1,000 laundering
15 currencies

→

01 Cleaning

Keep 1–10 Sep
Drop self-transfers
Convert amounts to USD

→

02 Features

25 point-in-time features:
payment, receiver (fan-in),
sender (fan-out)

→

03 Train XGBoost

Train 2–6 Sep (19.0 s, no GPU)
Cut-offs on 7–8 Sep (F2, F0.5)
Test once on 9–10 Sep

→

Export

Model + two cut-offs
to the shared folder
(~/shared/service_data)

2 SCORE EACH PAYMENT LIVE · Docker containers mule-service + studentshield on a private network

model + cut-offs loaded when mule-service starts

Payment lands

An incoming payment
to an account

→

mule-service /score

25 features → XGBoost score
under a millisecond
Top-3 SHAP reasons

→

Above student cut-off? (F0.5)

yes →

Student warning + analyst alert

Phone asks "Has someone asked you to move this money?"
Payments out are paused · 7 in 10 are real mule activity

no ↓

Above analyst cut-off? (F2)

yes →

Analyst alert only

Analyst sees account, money flow, reasons, rarity
Student's phone stays normal · 1 in 2 are real

no →

No action

Payment goes through

Student answers → analyst timeline

Re-asked only on new flagged money · after "Someone asked me to move it" or 3 answers, a specialist takes over

sends each payment

studentshield web app (port 5000)

The only public container
/demo real case, step by step
/live full test-day replay
Phone view + analyst console